

Bank Loan Analytics: Interview Cheat Sheet

Power BI + SQL + Python | 38,576 loans issued in 2021 | Problem → Data → Approach → Insights → Recommendation

60-SECOND PITCH

I built an end-to-end Bank Loan Analytics dashboard in Power BI on about 38,000 loans from 2021, to help a bank monitor lending performance and credit risk. I explored the data in Python, answered the business questions in SQL (CTEs, window functions), and built a two-page dashboard with DAX: page 1 shows KPIs with MTD and MoM change, page 2 breaks the portfolio down by month, state, term, purpose and home ownership. Main finding: about 14% of loans were charged off, with risk concentrated in lower grades, 60-month terms and small business loans.

NUMBERS TO REMEMBER

Applications	38,576	Avg interest rate	12.05%
Funded amount	\$435.8M	Avg DTI	13.33%
Amount received	\$473.1M	Good / Bad loans	86.18% / 13.82%
Bad loans repaid only	~57% (\$37.3M of \$65.5M)	Monthly apps	2,332 (Jan) to 4,314 (Dec)
Bad rate by grade	5.7% (A) to 31.3% (G)	Bad rate by term	60m 22.3% vs 36m 10.7%
Riskiest purpose	Small business 25.6%	Top state	CA: 17.9% of apps, \$78.5M

RECOMMENDATIONS (THE “SO WHAT”)

- Tighten approval or raise pricing for Grades E to G and 60-month loans; review small business criteria; monitor California (largest market) and Florida (17.3% bad rate, highest among top states).

LIKELY QUESTIONS AND SHORT ANSWERS

- Good vs bad loan? Good = Fully Paid or Current; Bad = Charged Off. Limitation: Current loans are unfinished, so the bad rate could rise.
- MTD and MoM? MTD = latest month's value. MoM = (MTD – previous month) ÷ previous month. SQL: LAG(). Power BI: DAX time intelligence (name the functions you used, e.g. TOTALMTD, DATEADD).
- Why SQL if you have Power BI? SQL is where I validated numbers and did heavy aggregation; computing KPIs in both tools let me cross-check the dashboard.
- Explain a window function. LAG(funded) OVER (ORDER BY month) returns the previous month's value in the same row, so I can calculate growth.
- Hardest part? Getting MoM right with a partial December; cleaning quirks (TERM has a leading space; INT_RATE and DTI stored as decimals).
- Higher rate = more defaults? Not necessarily. Higher rates go to riskier borrowers, so it may be the grade driving it. Association, not causation.
- What next? Default-prediction model (e.g. logistic regression), more years of data for seasonality, and profitability (interest earned vs charge-off losses).

HONEST CAVEATS (SAY THESE, IT BUILDS CREDIBILITY)

- Data covers 2021 only and ends 12 Dec, so December growth (+6.9% apps, +13.0% funded vs November) compares a part month with a full one.
- Descriptive analysis: it explains what happened, not what will happen. Small segments (e.g. “OTHER” home ownership, 98 loans) need caution.
- Be ready to explain every step and re-run a few queries live. Only claim what you did.